EWMAC Trading System: Operations and Strategy Manual

Amare Capital Management (Pty) Ltd | Proprietary Trading Firm | Pretoria, South Africa Covers: EWMAC_BOT v4.14, EWMAC_FORECAST v1.09, EWMAC_OVERLAY v1.05, and VALUE_MOMENTUM_SCREENER.py, DISCOUNTED_CASH_FLOW.py, and TRADINGVIEW_PINESCRIPT_EWMAC | Manual issued 10 October 2026

Internal use only. This manual describes how the software behaves as written. It does not promise returns. All parameter suggestions are starting points to be tested before live capital is used.

1. Purpose and scope

This manual explains how Amare Capital Management runs the EWMAC trend-following system end to end:

- what the strategy is and how the code calculates it,
- how to choose a pair, timeframe and trading style,
- how the ATR trailing stop works and how to configure it,
- every input on the EA and both indicators,
- how to host the system 24/7 on an AWS EC2 Windows server running MetaTrader 5, • how to monitor it from a phone using Microsoft's remote desktop client.

The version covered is the single-timeframe EA: one EA instance computes its forecast on one timeframe (SignalTimeframe) for the chart symbol it is attached to.

2. System components

File	Type	Role
EWMAC_BOT.mq5	Expert Advisor	Calculates the forecast, trades stop-and-reverse, optionally manages an ATR trailing stop, draws a dashboard
EWMAC_OVERLAY.mq5	Chart indicator	Draws fast and slow EWMA lines on price, colour-flips the fast line, prints buy/sell arrows
EWMAC_FORECAST.mq5	Sub-window indicator	Plots the capped forecast (-20 to +20) with signal arrows and a text HUD

The three files use the same maths, so what you see on the indicators is what the EA acts on, provided the settings match (see section 8.4).

3. The strategy: EWMAC

EWMAC means Exponentially Weighted Moving Average Crossover. It is a systematic trend-following rule of the kind used by managed-futures funds. The idea: when a faster average of price is above a slower average, the market is trending up; when it is below, trending down.

3.1 How the forecast is built

For every closed bar on the chosen timeframe:

1. Compute a fast EMA and a slow EMA of the close (smoothing alpha = 2 / (period + 1), seeded with a simple average).
2. Raw signal = fast EMA minus slow EMA.
3. Volatility = standard deviation of bar-to-bar close changes over StdevWindow bars (default 50).
4. Forecast = (raw signal / volatility) x scalar.
5. The forecast is capped to the range ForecastCapMin to ForecastCapMax (default -20 to +20).

Dividing by volatility makes the forecast comparable across symbols and calm or wild periods. The scalar rescales the result so the average absolute forecast is about 10. A forecast near +20 is a very strong uptrend; near 0 is no trend.

Illustration: fast minus slow = 1.2, volatility = 0.4, pair 8/32 (scalar 5.3). Forecast = 1.2 / 0.4 x 5.3 = 15.9, which the HUD labels STRONG BUY.

3.2 The six pairs and their scalars

Each pair has a slow period four times the fast period. Faster pairs react sooner and flip more often; slower pairs ride longer trends and flip less.

Pair (input)	Fast	Slow	Carver scalar	Alternative scalar (10 / sqrt(fast))
EWMAC_2_8	2	8	10.6	7.07
EWMAC_4_16	4	16	7.5	5.00
EWMAC_8_32	8	32	5.3	3.54
EWMAC_16_64	16	64	3.75	2.50
EWMAC_32_128	32	128	2.65	1.77
EWMAC_64_256	64	256	1.87	1.25

UseCarverScalar = true uses the middle column; false uses the last column.

3.3 What the EA actually trades on

The EA decides direction from the **sign** of the forecast only: above zero means long, below zero means short. Because the scalar, the volatility window and the caps are all positive rescalings, they change the number shown on the dashboard and indicators but **not** when the EA buys or sells. In EWMAC_BOT v4.14, the inputs that change trading behaviour are the pair, the signal timeframe, TradeCurrentTrend, lot size, the ATR trailing stop settings and execution settings. StdevWindow, the caps and the scalar choice are display and diagnostic settings for the EA. They would matter for position scaling if that is added later.

4. Choosing a pair and timeframe

The EMA periods count **bars**, so the same pair means a different time span on each timeframe. The table shows the real-time span of the fast and slow EMA.

Pair	M1	M5	M15	H1	H4
2/8	2m / 8m	10m / 40m	30m / 2h	2h / 8h	8h / 32h
4/16	4m / 16m	20m / 80m	1h / 4h	4h / 16h	16h / 64h
8/32	8m / 32m	40m / 2.7h	2h / 8h	8h / 32h	32h / 5.3d
16/64	16m / 64m	1.3h / 5.3h	4h / 16h	16h / 2.7d	2.7d / 10.7d
32/128	32m / 2.1h	2.7h / 10.7h	8h / 32h	32h / 5.3d	5.3d / 21d
64/256	1.1h / 4.3h	5.3h / 21h	16h / 2.7d	2.7d / 10.7d	10.7d / 43d

4.1 Style archetypes

These are illustrative starting points for testing, not validated settings.

Style	Timeframe	Pair	ATR trail	Character
Fast / scalping	M1 to M5	4/16 or 8/32	Off, or tight (1.5 x 14)	Many flips and many trades. Spread and commission costs dominate. Needs close monitoring.
Intraday	M15	8/32 or 16/64	Standard (2.0 x 14)	Holds from hours to a day or two. A balance of responsiveness and trade count.
Swing	H1 to H4	16/64 or 32/128	Wide (3.0 x 14 or 21)	Holds days. Fewer trades; whipsaw in ranges costs less per trade.

Position Investing / trend H4 to D1	H4 to D1	32/128 or 64/256	Very wide (3 to 4 x 20), or off	Few trades, long holds, large open profit swings. Needs a larger margin buffer.

Rules of thumb:

• Faster pair or lower timeframe = more trades = more cost drag and more whipsaw. • Slower pair or higher timeframe = later entries and exits, but fewer false flips.

5. EA behaviour in detail

5.1 Timing

The EA reads the forecast once per new bar on SignalTimeframe. It uses the last **closed** bar and the one before it, so a decision is made from completed data and does not flicker while a bar forms. The order is then sent on the first tick of the new bar.

5.2 Stop-and-reverse

- Forecast above zero: target is BUY. Below zero: target is SELL.
- When the target flips, the EA closes the opposite position and opens the new one. With the ATR trail off, the EA is always in the market once started.
- At most one position per symbol and MagicNumber is held. Positions from other magic numbers or manual trades are ignored by the EA's position logic.

5.3 TradeCurrentTrend

- **true (default):** on attach (or restart) the EA immediately enters in the direction of the current forecast.
- **false:** the EA stays flat until the forecast next crosses zero, then trades.

On a server restart or EA re-attach, the EA's memory resets, so with true it will re-enter at once. Set it deliberately.

5.4 Blocked side (wait for next cross)

If the EA sees its position disappear without it closing it (an ATR stop hit, a manual close, or a broker stop-out), it will not re-enter in the same direction. The dashboard shows the target as BUY (WAIT CROSS) or SELL (WAIT CROSS). Trading resumes at the next zero crossing. Use this on purpose: closing a position by hand keeps the EA out until the next signal.

5.5 Safety checks before an order

The EA will not send an order if: trading is not allowed in the terminal, the EA or the account; the symbol

cannot be selected; no tick is available; the lot size is invalid after rounding to the symbol's lot step and limits; or required margin exceeds free margin. Each block is written to the Experts log. The EA also validates inputs at start-up and refuses to run with, for example, StdevWindow below 2, WarmupBars below 100, a cap range that is inverted, non-positive lots, or an enabled ATR trail with a non-positive period or multiplier.

5.6 Multiple instances

Positions are matched by symbol **and** MagicNumber. Two EAs on the same symbol must have different MagicNumber values or they will treat each other's positions as their own. Different symbols can share a magic number, but unique values per instance are cleaner for reporting.

6. ATR trailing stop

ATR (Average True Range) measures typical bar range. The trail distance is ATR x ATR_Multiplier, so the stop widens in volatile markets and tightens in quiet ones. The ATR is calculated on SignalTimeframe with MT5's built-in ATR indicator.

6.1 How it works

1. **Initial stop at entry.** When UseATRTrail is true, the order is sent with a stop-loss at entry price minus (long) or plus (short) the trail distance, pushed out if needed to respect the broker's minimum stop distance.

2. **Trailing.** On each tick the EA computes a candidate stop from the current bid (long) or ask (short) minus or plus the trail distance. The stop only moves in the favourable direction, never back. 3. **Activation.** The stop begins to trail only once the candidate is better than the entry price. Until price has moved more than one trail distance in profit, the initial stop stays where it was placed. 4. **Stop hit.** The broker closes the trade at the stop. The EA then waits for the next zero cross (section 5.4).

5. **Zero cross before stop.** If the forecast flips first, normal stop-and-reverse applies and a new stop is set on the new trade.

Because the stop is a real order attached to the position on the broker server, it still protects the position if the cloud server or MT5 goes offline. The stop-and-reverse signal does not, since that is generated by the running EA.

6.2 Style presets (starting points)

Style	ATR_Period	ATR_Multiplier	Effect
Tight	7 to 10	1.0 to 1.5	Exits early on small pullbacks. More stop-outs and wait-for-cross periods.
Standard	14	2.0	The default. Moderate room for pullbacks.
Wide	14 to 21	3.0	Gives trends room; gives back more open profit before exit.

Very wide	20 to 30	4.0 or more	Close to a disaster stop; the EWMAC reversal is usually the real exit.

A larger ATR_Period smooths the distance; a smaller one reacts faster to volatility changes. Always match the stop style to the pair: a fast pair on a low timeframe suits tight stops, while a slow pair needs wide ones or the stop will cut trends the signal would have kept.

6.3 Visual styles of the trail line

With PlotTrailOnChart on, the current stop is drawn as a horizontal line, and the stop's recent path is drawn as connected segments.

Input	Default	What it does
TrailColor	Orange	Colour of the current stop line
TrailWidth	1	Thickness of the current stop line
TrailStyle	Solid	Line pattern: solid, dash, dot, dash-dot or dash-dot-dot
TrailHistoryColor	Orange	Colour of the stop's historical path
TrailHistoryWidth	1	Thickness of the path
TrailHistoryStyle	Dot	Line pattern of the path
TrailHistoryMaxPoints	100	Number of stop moves remembered before the oldest is dropped

A common choice is a solid bold line for the live stop and a thin dotted path for history, in colours that contrast with the chart background. Lines are removed when no position is open or when the trail is off.

7. EA input reference (EWMAC_BOT)

7.1 EWMAC Strategy

Input	Default	Meaning
EWMACPair	8/32	Fast and slow EMA periods (section 3.2)
SignalTimeframe	M1	Timeframe the forecast and ATR are computed on. Independent of the chart you attach to.
StdevWindow	50	Bars used for the volatility measure (display only for trades)

ForecastCapMin / Max	-20 / +20	Forecast limits (display only for trades)
WarmupBars	1000	History bars loaded to seed the EMAs. Minimum 100.
TradeCurrentTrend UseCarverScalar	true true	Enter at once in the current direction on attach Table scalar vs 10 / sqrt(fast)

WarmupBars: the EMA seed influence fades with each bar. At 1000 bars even the slowest pair has a negligible seed effect. The EA also enforces a floor of slow period + StdevWindow + 50 bars. Make sure the symbol has that much history on SignalTimeframe loaded in MT5.

7.2 Position Sizing, ATR, Execution, Notifications

Input	Default	Meaning
FixedLots	0.01	Volume per trade, rounded down to the symbol's lot step

Input	Default	Meaning
UseATRTrail	false	Enable the ATR trailing stop
ATR_Period	14	ATR length
ATR_Multiplier	2.0	Trail distance in ATRs
PlotTrailOnChart	true	Draw the stop lines
MagicNumber	20260101	Identifier for this EA's positions
SlippagePts	50	Maximum price deviation allowed, in symbol points
SendPush	true	Push notification to the MT5 mobile app on each trade
SendPopup	false	On-screen alert on each trade
ShowComment	true	Show the dashboard panel

Push notifications need your MetaQuotes ID entered in MT5 on the server (Tools > Options > Notifications). The ID is shown in the MT5 mobile app under Settings > Messages.

7.3 Dashboard

The panel at the top right of the chart shows: symbol, timeframe, pair, scalar, current and previous

forecast, trend (BULLISH, BEARISH, NEUTRAL), open position (FLAT, BUY, SELL), target (with WAIT CROSS when blocked), lots and ATR trail status with its period and multiplier. It adapts to light or dark chart backgrounds.

8. Indicators

8.1 EWMAC_OVERLAY (on the price chart)

- **Fast EWMA** line: green when fast is above slow, red when below.
- **Slow EWMA** line: blue.
- **Arrows:** green up-arrow under the candle at a buy cross, red down-arrow above the candle at a sell cross. Arrows appear only on closed bars, so they do not repaint.
- Inputs: Pair, StdevWindow, CapMin, CapMax, BuyThreshold, SellThreshold, UseCarverScalar, ArrowOffsetPts (arrow distance from the candle in points, default 5).

8.2 EWMAC_FORECAST (sub-window)

- Blue forecast line with level lines at 0, the cap values and the thresholds; vertical scale is the caps plus a margin of 5.
- Buy arrows at the bottom and sell arrows at the top of the pane (just beyond the caps), on closed bars only.
- A text HUD: pair, scalar, forecast value and a strength label.
- Extra inputs: ShowHUD and HUD_Y (vertical position).

HUD strength bands:

Forecast	Label
17.5 and above	VERY STRONG BUY
12.5 to 17.5	STRONG BUY
7.5 to 12.5	BUY
2.5 to 7.5	WEAK BUY
-2.5 to 2.5	NEUTRAL
-7.5 to -2.5	WEAK SELL
-12.5 to -7.5	SELL
-17.5 to -12.5	STRONG SELL
below -17.5	VERY STRONG SELL

The label is information only. The EA trades on the zero cross, not on these bands.

8.3 Thresholds

BuyThreshold and SellThreshold default to 0, meaning signals fire when the forecast crosses zero. They must satisfy: BuyThreshold is not below CapMin, SellThreshold is not above CapMax, and BuyThreshold is not below SellThreshold. A split such as +5 and -5 makes the indicator signal only on stronger moves, but the **EA always uses zero**, so keep 0 and 0 if the chart should show what the EA does.

8.4 Keeping indicators and EA in sync

- Attach the indicators to a chart whose timeframe equals the EA's SignalTimeframe. The indicators follow the chart timeframe; the EA follows its own input.
- Use the same Pair and UseCarverScalar. Keep both thresholds at 0.
- Minor differences at the left edge of the chart are expected, because the indicators compute from the whole history and the EA from WarmupBars.

9. Cloud hosting: AWS EC2 (free tier)

A forex or CFD EA must run continuously. A home PC sleeps, loses power and drops connection. An EC2 Windows server runs MT5 around the clock.

9.1 What the free tier covers

AWS's free offer has changed. Accounts created before 15 July 2025 can fall under the legacy tier: 750 hours a month of micro instances (t2.micro, or t3.micro where t2 is not offered) for the first 12 months, plus 30 GB of EBS storage. Newer accounts get a free plan based on credits for a limited period instead. Before relying on it, check your account's own entitlements in the AWS Billing console (Free Tier page), and create a **budget alert** so a surprise charge emails you. Public IPv4 addresses, extra storage, data transfer and instances above micro size can all incur charges.
A micro instance has 1 vCPU and 1 GB RAM. Windows Server plus MT5 fits only if you keep the setup lean: one MT5 terminal, a few charts, no other software. If the server becomes sluggish or memory is exhausted, move up to a larger instance size, which is no longer free.

9.2 Launch the server

1. Sign in to the AWS console and choose a **region**. Pick one close to your broker's trade server to cut latency (check the ping figure in the bottom right of MT5 afterwards). Some regions, such as Cape Town, must be enabled in account settings first.
2. EC2 > Launch instance. Name it, for example AMARE-EWMAC-01.
3. Image: **Windows Server 2022 Base** (or the current Base image). Choose an instance type marked Free tier eligible.
4. Key pair: create one and store the .pem file safely. It is needed to decrypt the Windows password.
5. Network: allow inbound **RDP (TCP 3389)** only from your own IP address. Do not use 0.0.0.0/0. See section 11 for mobile networks.
6. Storage: 30 GB gp3 is enough.
7. Launch. When the status checks pass, select the instance > Connect > RDP client > Get password,

upload the key and decrypt.

8. Note the public IPv4 address. It changes on stop and start unless you attach an Elastic IP; check the current charge for IP addresses before doing so.

9.3 Prepare Windows

1. Connect with Microsoft Remote Desktop from your PC using Administrator and the decrypted password. Change the password immediately.
2. Run Windows Update, then set active hours so automatic restarts do not land during market hours.
3. Set the server time zone as you prefer; MT5 uses broker server time for bars regardless.
4. Avoid installing browsers extensions, games or heavy tools on this machine.

9.4 Reboots and persistence

- **Close the remote desktop window or choose Disconnect. Do not Sign out.** A disconnected session keeps MT5 and the EA running; signing out ends them.
- After an unattended reboot, MT5 does not start until someone signs in unless you configure automatic sign-in, which carries a security trade-off. Plan to reconnect and restart MT5 after Windows updates.
- Remember that restarting MT5 re-attaches the EA with a fresh memory (section 5.3). • A single instance running all month uses about 744 of the 750 free hours. A second instance would exceed the allowance.

10. MetaTrader 5 setup

1. Download MT5 from your broker (Amare's account: Exness, ZAR-denominated) and install it on the server.
2. File > Login to Trade Account and enter the account number, password and server. Use the trading password, not the investor password, or the EA cannot place orders.
3. Market Watch: right-click > Symbols, and enable the instruments you trade. 4. Tools > Options > Expert Advisors: tick **Allow algorithmic trading**. Leave the toolbar **Algo Trading** button green.
5. File > Open Data Folder. Copy EWMAC_BOT.mq5 into MQL5\Experts and the two indicators into MQL5\Indicators.
6. Open MetaEditor (F4), open each file and compile (F7). Confirm 0 errors. Back in MT5, refresh the Navigator.
7. Open a chart for the symbol. Set the chart timeframe to your intended SignalTimeframe. Add EWMAC_OVERLAY and EWMAC_FORECAST with the same pair.
8. Drag EWMAC_BOT onto the chart. On the **Common** tab tick Allow Algo Trading. On the **Inputs** tab set pair, SignalTimeframe, FixedLots, ATR options and a unique MagicNumber. OK. 9. Confirm a smiley face at the top right of the chart and the dashboard in the top right corner. Check the Experts tab of the Toolbox for the start-up banner.
10. Save the layout as a profile (File > Profiles > Save As) so charts return after a restart.

First-run test: use a demo account and the minimum lot size. Confirm a trade opens in the forecast's direction, that a stop appears when ATR is on, and that push notifications arrive on your phone.

11. Mobile access with Microsoft's remote desktop client

Microsoft has replaced its Remote Desktop app on iPhone, iPad and Android with the **Windows App**; on some Android phones the older Remote Desktop app may still be installed. Both connect the same way. Check the app store for the current listing.

11.1 Connect

1. Install the app on your phone.
2. Add a PC (or Remote PC) and enter the server's public IPv4 address or public DNS name. 3. Set the user account to Administrator and enter the password. Accept the certificate prompt the first time.
4. Connect. Use touch controls to click and pinch to zoom; use the keyboard button for text entry.
5. When finished, **disconnect**, never sign out.

11.2 Mobile network access

A phone's IP address changes between Wi-Fi and mobile data. If the security group only allows your home IP, the phone will be refused. Options, from safest: use AWS Systems Manager Fleet Manager or a VPN with a fixed address, add the phone's current IP to the security group as needed and remove it afterwards, or allow a narrow range from your mobile provider. Never leave port 3389 open to the whole internet, as exposed RDP is a common attack target. Use a long unique Administrator password.

11.3 What to do from the phone

• **Monitor:** check the dashboard, open position, stop line and the Experts log. • **Intervene sparingly:** a manual close makes the EA wait for the next cross (section 5.4). To stop trading entirely, switch off the Algo Trading button.

- For routine checks, the MT5 mobile app signed in to the same account shows positions and receives the EA's push alerts with less effort than a remote session. Closing a trade there also triggers the wait-for-cross behaviour.

12. Daily operating procedure

Morning check (two minutes):

- Server reachable; MT5 shows a connected status and a normal ping.
- Algo Trading button green; smiley face on each EA chart.
- Dashboard forecast and position agree with the indicator pane; target is not stuck at WAIT CROSS unless expected.
- Experts log shows no repeated BLOCKED or ORDER FAILED lines.
- Free margin comfortably above the required margin; equity matches expectations.

Weekly: apply Windows updates at a planned time, review closed trades against signals, confirm free-tier usage and billing alerts, back up the .mq5 files and the saved profile.

13. Fundamental Analysis (INVESTING)

13.1 Value and Momentum Screener

Value_momentum_screener.py Ranks JSE tickers on valuation and price momentum, then combines the two into a signal. It saves the result to JSE_Stock_Rankings_<timestamp>.csv.

1. Data: pulls 1 year of prices (at least 126 days or the ticker is skipped) plus valuation fields from Yahoo Finance. Prices are converted from cents to rand.
2. Value score:
    - Uses P/E, P/B, P/S, EV/EBIT and EV/Gross Profit.
    - Missing values are filled with the cross-sectional median.
    - Each multiple is percentile-ranked, with lower multiples scoring higher. The five scores are averaged.
3. Momentum score: percentile-ranks the 1M, 3M, 6M and 12M returns and averages them equally. The 12M return falls back to the full-period return if there are fewer than 252 days.
4. Composite: 50% value + 50% momentum.
5. Signal: by composite percentile: top 20% STRONG BUY, 60–80% BUY, 40–60% HOLD, bottom 40% SELL.

13.2 Discounted Cash Flow Model

Values JSE stocks using Yahoo Finance annual statements and compares the result to the current price.

1. Data: pulls the income statement, balance sheet, cash flow, price (cents converted to rand) and beta. Non-ZAR statements are converted at the latest FX rate.
2. FCFF: NOPAT (EBIT × (1 − effective tax)) + depreciation + capex + working capital change. If that comes out ≤ 0, it uses a net income based fallback. Effective tax is capped at 0–60%, defaulting to 27%.
3. Discount rate (WACC):
    - Cost of equity is CAPM: x% risk-free + beta × (x% − y%).
    - Cost of debt is interest expense ÷ debt, after tax.
    - Weights use market cap against total debt, and WACC is clamped to x–y%.
4. Projection: grows FCFF at x% for y years and discounts each year. Terminal value uses x% perpetual growth (Gordon growth), discounted back.
5. Fair value per share: (PV of FCFFs + PV of terminal value + cash − debt) ÷ shares.
6. Output per company:
    - Verdict (undervalued or overvalued), upside %, margin of safety, and a ±20% fair value band.
    - FCFF, terminal value, equity value, WACC, ROIC and excess returns over WACC, beta and tax rate.
    - A warning if FCFF ≤ 0.

All reports go into one text file

14. Trading View EWMAC indicator (Pine Script v6)

A Carver-style trend forecast, plotted in its own pane. The fast and slow EMAs are also drawn on the price

chart.

1. **Pair:** a dropdown of 2/8, 4/16, 8/32, 16/64, 32/128 and 64/256, default 16/64. Each pair has its Carver scalar (10.6, 7.5, 5.3, 3.75, 2.65, 1.87).
2. **Forecast:**
    - Raw signal is fast EMA − slow EMA.
    - It is divided by the rolling stdev of price changes (default window 1440) and multiplied by the scalar.
    - The result is capped to ±20.
3. **Stdev fallback:** if stdev is below 0.1 × the minimum tick, or there are fewer than 1440 bars of history, it uses 0.2% of price.
4. **Signal label:** the capped forecast is mapped to Carver's Table 16 opinion scale:
    - ≥17.5 is Very strong buy and ≥12.5 is Strong buy.
    - ≥7.5 is Buy and ≥2.5 is Weak buy.
    - Between −2.5 and 2.5 is Neutral.
    - Below that are the mirrored sell labels, down to Very strong sell at −17.5 or lower.
5. **Buy/sell events:** a buy fires when the forecast crosses above the buy threshold (default 0), and a sell when it crosses below the sell threshold (default 0). These tint the background.
6. **Visuals:**
    - The forecast is a line filled green above the buy threshold and red below the sell threshold.
    - Dashed lines mark the thresholds and dotted lines mark the caps.
    - On the price chart, the fast EMA is green when above the slow EMA and red when below, with a faint background tint on each EMA cross.
7. **HUD table:** top right, showing the pair, scalar, capped forecast and signal label.

Differences from the EA

- **Stdev window:** 1440 here against 50 in the EA, so forecast magnitudes will differ.
- **Default pair:** 16/64 here against 8/32.
- **EMA seeding:** Pine's `ta.ema` seeds from the first bar, while the EA seeds with an SMA. Values converge after enough bars.
- **Sign:** the sign and zero-cross timing should match. The magnitudes are what diverge.
- **No trade logic:** the indicator has no entries, exits or ATR stop.

15. Risk controls and known limitations

- **Trend following loses in ranging markets.** Repeated small losses from whipsaws are normal and are the price of catching large trends. Expect long flat or losing stretches.
- **Costs matter.** Faster pairs and lower timeframes trade more. Spreads, swaps and commissions on CFDs can overwhelm a fast configuration.
- **Fixed lot size only.** The EA does not scale size by forecast strength or volatility, and has no daily loss limit, drawdown stop, spread filter, session filter or news filter. Add risk limits at the account and portfolio level yourself.
- **Always in the market without ATR.** A pure stop-and-reverse position can lose heavily on gaps

and fast moves. Leverage and margin can amplify this; size conservatively.

• **Slippage and gaps.** Stops and market orders can fill away from the intended price. • **Single point of control.** One EA per symbol and magic number; a server outage stops new signals, though broker-side stops remain.

• **Backtest before live.** Use the MT5 Strategy Tester with real-tick data and your broker's spreads, then forward-test on demo. Past results do not guarantee future performance.

Appendix A: Quick-start configurations

Copy the setting, attach the matching indicator pair on the same timeframe and test on demo first.

Name	SignalTimeframe	Pair	ATR	Notes
Fast	M5	8/32	1.5 x 14	Higher turnover, watch costs
Intraday	M15	8/32	2.0 x 14	Balanced starting point
Swing	H1	16/64	3.0 x 14	Fewer trades
Trend	H4	32/128	3.0 x 20	Few trades, larger swings
Pure reverse	any	any	Off	Always in market

Appendix B: Glossary

EMA / EWMA: exponentially weighted moving average, which weights recent prices more heavily. **Forecast:** the volatility-scaled, capped EWMAC value. **Scalar:** multiplier that normalises forecast size.

Stop-and-reverse: closing one direction and immediately opening the other. **ATR:** average true range, a volatility measure. **Magic number:** the ID an EA stamps on its trades. **Algo Trading:** MT5's master switch for automated trading. **RDP:** Remote Desktop Protocol, used to reach the cloud server.